5. Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

CODING:

```
import math


X = [[1, 2], [2, 3], [3, 3],
    [6, 7], [7, 8], [8, 7]]
Y = ['A', 'A', 'A', 'B', 'B', 'B']


def distance(a, b):
   return math.sqrt(sum((a[i] - b[i]) ** 2
                        for i in range(len(a))))


def knn(sample, k):
   distances = []
   for i in range(len(X)):
      d = distance(sample, X[i])
      distances.append((d, Y[i]))

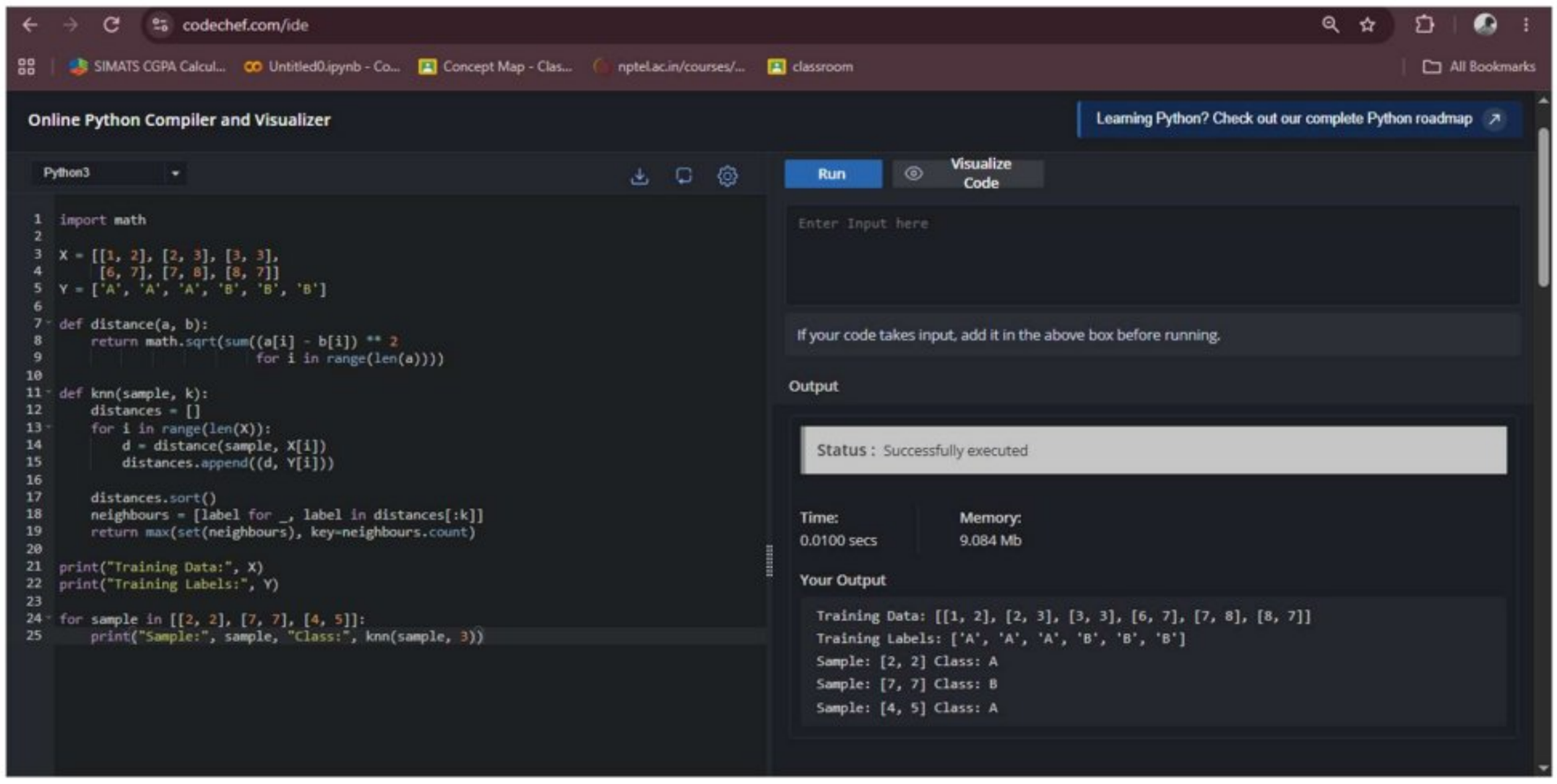

   distances.sort()
   neighbours = [label for _, label in distances[:k]]
   return max(set(neighbours), key=neighbours.count)


print("Training Data:", X)
print("Training Labels:", Y)


for sample in [[2, 2], [7, 7], [4, 5]]:
   print("Sample:", sample, "Class:", knn(sample, 3))
```

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OUTPUT:




```python
import math

X = [[1, 2], [2, 3], [3, 3],
     [6, 7], [7, 8], [8, 7]]
Y = ['A', 'A', 'A', 'B', 'B', 'B']

def distance(a, b):
    return math.sqrt(sum((a[i] - b[i]) ** 2
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def knn(sample, k):
    distances = []
    for i in range(len(X)):
        d = distance(sample, X[i])
        distances.append((d, Y[i]))

    distances.sort()
    neighbours = [label for _, label in distances[:k]]
    return max(set(neighbours), key=neighbours.count)

print("Training Data:", X)
print("Training Labels:", Y)

for sample in [[2, 2], [7, 7], [4, 5]]:
    print("Sample:", sample, "Class:", knn(sample, 3))
```

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